

Introduction to Computer Simulation II

Homework 1

Due: Tuesday, 14 April 2026

1. Binder parameter for the 2D Ising model

Use one of your Monte Carlo simulation programs with local updates (Metropolis, Glauber, ...) to determine for the 2D Ising model the Binder parameter

$$U = 1 - \frac{\langle m^4 \rangle}{3\langle m^2 \rangle^2}$$

with $m = (1/L^2) \sum_{i=1}^L \sum_{j=1}^L s_{i,j}$ around the critical temperature $T_c = 2/\ln(1 + \sqrt{2}) = 2.269185\dots$. Consider square lattices of linear extent $L = 8, 16, 32$, and 64 with periodic boundary conditions. Plot this data as a function of temperature and convince yourself that the curves for different lattice sizes cross quite accurately around T_c . Which value does U take at the crossing point?

2. Susceptibility of the 2D Ising model

Use one of your Monte Carlo simulation programs with local updates to determine for the 2D Ising model the susceptibility

$$\chi = \beta V (\langle m^2 \rangle - \langle |m| \rangle^2)$$

with $m = (1/V) \sum_{i=1}^L \sum_{j=1}^L s_{i,j}$, $V = L^2$ around the critical temperature $T_c = 2/\ln(1 + \sqrt{2}) = 2.269185\dots$. As in homework 1, consider square lattices of linear extent $L = 8, 16, 32$, and 64 with periodic boundary conditions. Determine the maxima of χ and test the finite-size scaling ansatz $\chi_{\max} \propto L^{\gamma/\nu}$.